Frequentist Approximate Sufficiency of Spectral Information for SBM Partitions

Spectral-LRT research note

12 October 2026

Abstract

The partition C is a fixed parameter throughout this note. No prior, random partition, or posterior argument is used. Uniform exact recovery from a statistic implies approximate sufficiency for bounded finite-action decisions whose loss depends only on the partition, even when the connection matrix is unknown. Combining this reduction with a published rowwise eigenspace perturbation theorem proves the result for every fixed number of blocks, uniform spectral separation, and $np \geq M \log n$ with a sufficiently large constant M. A separate fixed-known-P result gives ordinary Le Cam asymptotic sufficiency. These statements do not assert equivalence of the full experiment with unknown parameter (C, P).

1 Fixed-parameter model and the target notion

Let C be a partition of $[n]$ into K nonempty blocks. Label permutations are identified. A canonical ordering, for example by the smallest vertex in each block, supplies indices for the probability matrix P. Its indices must always be changed together with any alternative block ordering. For each fixed $\theta = (C, P)$, independently for $i < j$,

$$A_{ij} \sim \mathrm{Ber}(P_{c_i c_j}), \qquad A_{ji} = A_{ij}, \qquad A_{ii} = 0.$$

The symmetric matrix P contains the actual probabilities. Set $p = \max_{a,b} P_{ab}$. Fix $K \geq 2$ and positive constants π_0, κ, η. The parameter class satisfies

$$n_a \geq \pi_0 n, \qquad \kappa p \leq P_{ab} \leq p \leq 1 - \eta, \qquad \sigma_{\min}(P) \geq \kappa p. \tag{1}$$

There is no convergence assumption on P/p or n_a/n. Retain the K adjacency eigenpairs with largest absolute eigenvalues, including their signs:

$$T(A) = (U, \Lambda), \qquad U^\top U = I_K.$$

Use measurable deterministic tie conventions. Write $\mathsf{P}_{C,P}$ for the graph distribution and $\mathsf{Q}_{C,P}$ for the induced distribution of T. Neither is a distribution placed on the parameter.

For a finite action space $\mathcal{D}$ and any loss $0 \leq L(C, d) \leq 1$, let $R_{C,P}(\delta) = \mathbb{E}_{C,P} L(C, \delta)$. We say that T is *ϵ-sufficient for partition decisions* if, for every such loss and every possibly randomized full-data rule δ_A, there is a rule δ_T using only T such that

$$\sup_{(C,P) \in \Theta_n} \{R_{C,P}(\delta_T) - R_{C,P}(\delta_A)\} \leq \epsilon. \tag{2}$$

All losses here are invariant under block-label permutations and evaluate the partition only. This is the ordinary decision-theoretic comparison restricted to the parameter of interest. The rule δ_T cannot use the unknown P.

2 A finite-sample reduction requiring no prior

Theorem 1 (Recovery implies partition-decision sufficiency)**.** *Suppose a statistic-based estimator satisfies*

$$\sup_{(C,P)\in\Theta_n} \mathsf{P}_{C,P}\{\widehat{C}(T) \neq C\} \leq \epsilon_n.$$

Then (2) *holds with* $\epsilon = \epsilon_n$.

Proof. For the given loss, choose a deterministic minimizer $d^*(c) \in \arg\min_{d\in\mathcal{D}} L(c,d)$, and use $\delta_T(t) = d^*(\widehat{C}(t))$. For each fixed (C,P),

$$\begin{aligned} R_{C,P}(\delta_T) &\leq \min_d L(C,d) + \mathsf{P}_{C,P}(\widehat{C} \neq C) \\ &\leq \min_d L(C,d) + \epsilon_n \\ &\leq R_{C,P}(\delta_A) + \epsilon_n. \end{aligned}$$

On the correct-recovery event the action is optimal for the fixed true partition; on its complement the additional loss is at most one. Every full-data action has loss at least the oracle minimum. Take the supremum over the fixed parameters. There is no averaging over C. □

3 A general fixed-K spectral premise

Theorem 2 (Uniform spectral exact recovery)**.** *For every $q > 0$ there are M_q, n_0, depending only on $K, \pi_0, \kappa, \eta, q$, such that under* (1)*,*

$$n \geq n_0, \quad np \geq M_q \log n \quad \Longrightarrow \quad \sup_{(C,P)\in\Theta_n} \mathsf{P}_{C,P}\{\widehat{C}(T) \neq C\} \leq n^{-q}.$$

The estimator described below needs neither P nor the population centers.

Population geometry. Let Z be the membership indicator, $N = \mathrm{diag}(n_a)$, $F = ZN^{-1/2}$, and $D_{ii} = P_{c_i c_i}$. The actual loop-free mean is

$$M = \mathbb{E}_{C,P} A = ZPZ^\top - D.$$

The matrix D is constant within each block. Consequently $\mathrm{col}(F)$ is an invariant subspace of M, and

$$F^\top M F = N^{1/2} P N^{1/2} - \mathrm{diag}(P_{aa}).$$

Its smallest singular value is at least $\pi_0 \kappa np - p$. On the orthogonal complement the eigenvalues are $-P_{aa}$, each with multiplicity $n_a - 1$. Thus, eventually, the K signal eigenvalues have absolute magnitude and separation from the remaining spectrum at least cnp, with an upper bound Cnp. Both signs are allowed.

If V diagonalizes the displayed K-dimensional restriction, the signal eigenvectors are $U_* = FV$. Their community centers obey

$$m_a = V_{a,:}/\sqrt{n_a}, \qquad \|m_a - m_b\|_2 = \sqrt{n_a^{-1} + n_b^{-1}} \geq \sqrt{2/n}, \qquad \|U_*\|_{2\to\infty} \leq (\pi_0 n)^{-1/2}. \quad (3)$$

This uses the full rank-K signal subspace. It would not follow by retaining an arbitrary smaller number of directions.

Published stochastic input and its specialization. Use Lei's generic Bernoulli eigenspace perturbation theorem [1, Theorem 3.4]. Apply it separately to the positive and negative signal tails, which are consecutive eigenvalue intervals, and combine their orthogonal alignments. Theorem 5.2 of that paper is a direct fixed-shape SBM recovery result; we instead specialize the generic bound to obtain uniform constants on (1).

The quantities in the generic theorem satisfy, uniformly,

$$\lambda_{\min}^{\mathrm{abs}}, \Delta \geq cnp, \quad p^*, \bar{p}^* \leq p, \quad \|M\|_{2\to\infty} \leq p\sqrt{n}, \quad \|U_*\|_{2\to\infty} \leq (\pi_0 n)^{-1/2}, \quad r \leq K.$$

Choose its confidence parameter $\delta = n^{-(q+2)}$ and auxiliary exponent $1/2$. Then $R = O(\log n)$ and $g = O(\sqrt{np} + \log n/\log\log n)$. Substitution in that theorem gives

$$\inf_{W\in O(K)} \|UW - U_*\|_{2\to\infty} \leq \frac{B_q}{\sqrt{n}}\sqrt{\frac{\log n}{np}}, \tag{4}$$

outside an event of probability at most n^{-q}, after increasing n_0. The intermediate terms are bounded by

$$\frac{B_q}{\sqrt{n}}\left\{\frac{1}{\sqrt{np}} + \frac{\log n}{np} + \sqrt{\frac{\log n}{np}}\right\}.$$

The additional factor

$$\frac{\sqrt{np}\, R^{1/4}}{np\log R}$$

is bounded and vanishes. For the final minimum in the generic theorem, use its mean-row-norm branch. This contributes at most a constant times $n^{-1/2}(np)^{-1/2}$. For $np \geq M_q \log n$ these terms yield (4). The theorem's gap requirement holds eventually. The accompanying operator-norm control identifies the retained empirical eigenpairs with the signal tails. No extra assumption from its Corollary 3.6 is needed. Constants depend only on the fixed bounds in (1).

Explicit decoder and completion of the proof. Construct an auxiliary graph on the vertices by connecting i, j if

$$\|U_{i,:} - U_{j,:}\|_2 \leq \frac{1}{2\sqrt{n}}.$$

Return its components if there are exactly K and each has size at least $\pi_0 n$. Otherwise return a fixed admissible fallback partition. The balance bound is a specified model-class constant, not unknown P. This is a measurable function of U; pairwise distances are unchanged by orthogonal alignment.

Increase M_q so that the right side of (4) is at most $1/(8\sqrt{n})$. On that event, within-community distances are at most $1/(4\sqrt{n})$, while between-community distances are at least $(\sqrt{2} - 1/4)/\sqrt{n} > 1/(2\sqrt{n})$ by (3). Every true block is therefore a clique and there are no cross-block edges. The components are exactly C. This proves Theorem 2.

Corollary 1 (Unknown-P partition sufficiency)**.** *Under Theorem 2, the spectral statistic is n^{-q}-sufficient for all bounded finite-action partition decisions. In particular (2) tends to zero uniformly with P unknown.*

Proof. Apply Theorem 1 with $\epsilon_n = n^{-q}$. □

4 Ordinary asymptotic sufficiency with fixed known P

This section treats C as the sole experiment parameter. The fixed matrix P is known to the statistician. Block indices must be identifiable under the convention used to generate the model: use the canonical partition order above, or assume correctly named-label recovery. Unlabeled recovery alone cannot justify reconstructing a model with misassigned named rows of P.

Theorem 3 (Known-P experiment equivalence)**.** *For a fixed-known-P grouping experiment, suppose* $\sup_C \mathsf{P}_{C,P}(\widehat{C}(T) \neq C) \leq \epsilon_n$*. Then the Le Cam distance between observing A and observing T is at most ϵ_n.*

Proof. Define the Markov kernel

$$\mathcal{K}_P(t, \cdot) = \mathsf{P}_{\widehat{C}(t),P}(\cdot).$$

It independently generates a new graph from the estimated partition and the known matrix P. The kernel does not depend on the unknown C. For each fixed C, with $\epsilon_C = \mathsf{P}_{C,P}(\widehat{C} \neq C)$,

$$\mathcal{K}_P \mathsf{Q}_{C,P} = (1 - \epsilon_C)\mathsf{P}_{C,P} + \epsilon_C \mathsf{H}_C$$

for some probability measure H_C (arbitrary when $\epsilon_C = 0$). Hence

$$\sup_C \|\mathcal{K}_P \mathsf{Q}_{C,P} - \mathsf{P}_{C,P}\|_{\mathrm{TV}} \leq \epsilon_n.$$

The other direction has deficiency zero because T is a deterministic function of A. Taking the maximum of the two deficiencies proves the claim. This is the standard experiment-comparison notion of asymptotic sufficiency [2]. No randomization of the parameter has occurred. □

5 Connection to the optimal error exponent

The following is an implication, not a new verification of an existing end-to-end decoding theorem. If a spectral-only estimator is independently proved to obey, uniformly over a parameter class,

$$\mathbb{E}_{C,P}\mathrm{Mis}(\widehat{C}, C) \leq \exp\{-(1 - o(1))I_n\},$$

then, since any wrong partition has misclassification at least $1/n$,

$$\mathsf{P}_{C,P}(\widehat{C} \neq C) \leq n\mathbb{E}_{C,P}\mathrm{Mis} \leq n \exp\{-(1 - o(1))I_n\}.$$

If $I_n \geq (1 + \tau) \log n$ for fixed $\tau > 0$, this tends uniformly to zero and gives the same sufficiency consequences. The existing manuscript's entire optimal-exponent argument is not independently re-certified here.

6 Precisely what is proved

The proved result covers every fixed K, balanced positive block sizes, uniform spectral separation, and logarithmic or denser degree with a sufficiently large coefficient. It requires no prior on C, no known P for partition decisions, no convergence of normalized model matrices, and no restriction to assortative or two-block graphs.

The result does not provide a sharp recovery constant, or sufficiency below the exact-recovery regime. Exact recovery is sufficient for our reduction; we do not assert that it is necessary for sufficiency itself. With unknown P the full (C, P) experiment is not declared equivalent: the kernel in Theorem 3 would then use an unknown parameter and be inadmissible. Finally the theorem concerns the information in T. It does not identify a fractional Bernoulli working likelihood with the true spectral density, prove convergence of a specific likelihood refinement, or establish MCMC mixing.

References

[1] Lihua Lei (2020). *Unified $\ell_{2\to\infty}$ Eigenspace Perturbation Theory for Symmetric Random Matrices*. arXiv:1909.04798v2, Theorems 3.4 and 5.2. https://arxiv.org/pdf/1909.04798.

[2] Yanjun Han (2021). *Statistical decision theory: model distance and equivalence*, Lecture 4, pp. 11–18. https://web.stanford.edu/class/ee378c/lecture4_annotated.pdf.